

Cell free RNA Detection of Pancreatic Cancer in Pre Diagnostic High Risk and Symptomatic Patients

Corresponding Author: Professor Thuy Ngo

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors identified 39 potential cfRNA biomarkers for the diagnosis of PDAC. They combined these biomarkers with a novel normalization method yielding discrimination accuracy for the separation of PDAC from all other intermediary pathologic conditions with an AUC of 0.966 by cross-validation in a discovery cohort and 0.889 in the independent validation cohort. The findings could help to identify high-risk individuals for direction to an EUS-FNA procedure thereby enhancing its diagnostic utility. The following are my concerns:

Line 128, "There were no statistically significant differences between the compositions of the sample sets (Table 1). In the PDAC groups, there were more males than females and the age range was higher than patients with other diagnoses ($P < 0.001$).". It would be beneficial if the authors could list all statistical p values for compositions of the sample sets in Table 1. I wonder will the significant gender and age range imbalance impact all the follow up analyses in this study? Have your models being adjusted for gender and age range?

L147, "To create this set of cfRNA intrinsic genes, we generated cfRNA sequencing data for plasma samples from three healthy individuals.". I wonder how did the author select the 3 healthy individuals? What the races are these 3 individuals? Moreover, the samples size here is very small and may not represent the broad healthy individuals to select the reference cfRNA genes for robust normalization.

L235, "To address the large imbalance between these two classes in the data, we used the SMOTE interpolation method [47] to create synthetic PDAC samples for training." I am concerned about the synthetic PDAC samples for training, how many synthetic PDAC samples were generated for training? Will the synthetic PDAC samples being too ideal and significantly different with the real PDAC samples? Moreover, it would be helpful to perform the training without synthetic PDAC samples as a comparison.

When normalizing gene expression across all samples, including pancreatic and other cancer samples, will the genes specifically expressed in the pancreas (e.g., CTRB1/2) be removed or normalized along with the genes?

(Remarks on code availability)

To make sure the paper are reproducible and the code is a reusable resource for the community, the authors need to add the tutorial/manual on how to use the codes, for instance, 2024_ngolab_cfRNA/Analysis_Scripts/DE_wilcox.R, what is the format of input file, the commands on how to use the codes, what is the output and how to interpret them?

Reviewer #2

(Remarks to the Author)

The manuscript entitled "Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-diagnostic High-risk and De-novo Symptomatic Patients" by Moore, Spiliotopoulos, Callahan et al. reports on a method and a 39 cell-free RNA gene classifier to diagnose pancreatic ductal adenocarcinoma in a pre-diagnostic high-risk de novo symptomatic patient populations. The manuscript is well written, the figures are illustrative, the results support the conclusions, and the study has

a promising impact for better future clinical management of patients. Nevertheless, I have a series of technical questions and comments to ascertain the validity of the results.

major comments:

1. The blood collection tube type and time between blood draw and plasma preparation are two important variables that impact the extracellular RNA repertoire. These variables, as well as others (e.g., fasting status at the time of blood draw, etc.), are not mentioned in the study but are important for replicating the study and evaluating potential confounding variables. According to recent guidelines, the authors should report pre-analytical variables, e.g., CEN/TS 17742:2022 ("Molecular in vitro diagnostic examinations – specifications for pre-examination processes for venous whole blood – isolated circulating cell free RNA from plasma").
2. While the authors make a laudable effort to remove variable genes that result from variable platelet content (because of different centrifugation protocols), I'm surprised they only looked at centrifugation to determine so-called batch effects and improve the normalization. It is well-known that the time between blood draw and plasma preparation is likely an even more important factor, which the authors did not evaluate here. This may be acceptable if this pre-analytical variable was controlled well in this study, but that information is not provided (see previous point). Anyhow, this variable's impact should at least be discussed and serve as a warning for the reader. The same is true for the level of hemolysis, the temperature during blood storage/transport before plasma preparation, etc.
3. This reviewer thought that you could only do one normalization procedure (defined as applying a sample-specific factor to remove technical variability based on sample-specific gene expression values). Therefore, I'm confused by the authors' application of their intrinsic normalization procedure followed by TMM normalization. Can the authors confirm that intrinsic followed by TMM differs from doing TMM only (when using the same set of genes)?
4. The raw sequencing data and data-processing and analysis scripts are publicly available. But is the patient annotation file also available (with matched identifiers between the raw data and the annotation)? Also, key intermediate result matrices should also be made available on the GitHub, like raw and normalized gene counts.
5. ROC AUC values indicate that the measurements have classification potential but are not very informative for the individual patient. The authors are invited to mention positive and negative predictive values (like they do for the CA19-9 biomarker).
6. I'm well aware of (often ignored or underappreciated) batch effects in RNA seq studies, resulting from RNA extraction and/or library preparation and/or sequencing lane/flow cell. It would be good if the authors could annotate their samples in terms of at least these potential batch effects. For RNA extraction, the authors mention randomization (to avoid batch effects), but what about library prep and sequencing? As an example, did all samples receive a unique index so all could be pooled and sequenced on the same set of sequencing lanes?
7. Given the pervasive nature of batch effects, I'm surprised that a sequencing-based classifier works so well when trained on one cohort/batch and tested on another one. Could the authors clarify how they ensured that the normalized results have the same scale and variance, assuming a RF classifier is sensitive to scale and variance? Simply following the same data processing pipeline is likely not going to result in comparable datasets. Did the authors ensure that the sequencing depth was the same between the 2 cohorts, was some additional data rescaling done after normalization, ...?
8. The use of a Kaplan-Meier plot forces the authors to define a cut-off that may not be relevant or correct. I suggest they do Cox proportional hazard modeling of the score instead, either with or without the inclusion of additional variables (like stage 1-2 vs 3-4, CA19-9, ...).
9. The methods section should provide more details: how much plasma was used as input for RNA extraction, the RNA elution volume, how much RNA was DNase treated, the RNA eluate volume after the second purification, library QC, library concentration measurement, library loading concentration on the sequencer, read length, how many genes were removed in the variance filter, how many remained after that filter, 'divided by the total expression of the top 100 intrinsic genes' (how is this done, what is 'total expression'), 'normalized across tissues using z-scores' (how is this done, is this really normalization?), software version are lacking, etc.
- 10.

minor comments:

1. Figure 1: What is benign pancreas? Is this a normal pancreas, or benign pancreatic cysts, or something else?
2. Line 140: Is 'batch' factor a good term? Plasma preparation differences mainly impact an individual sample, while 'batch' is a term that is typically used for systematic technical differences introduced for a set of samples that are processed together.
3. The batch factor is likely driven entirely by platelet RNA content. Can the authors demonstrate enrichment of platelet signal in the 'batch' genes?
4. Please replace all instances of housekeeping gene by reference gene.
5. Line 157: What is 'the total gene expression'?
6. As the authors measure extracellular RNA in blood plasma, the term 'expression' is not very well suited. Plasma does not express any transcript; instead, transcript are more or less (differentially) abundant, or show differences in concentration (but not in expression levels). Expression is a term better reserved for cellular RNA, as part of an active process.
7. The list of 100 intrinsic genes along with their TPM values should be given in a supplementary file.
8. Line 181: It is unclear if the average depth of ~1 million reads per sample is after deduplication.
9. Please be more precise about the statistical test used to identify the differentially expressed genes. A Wilcox test is not a test. Do the authors mean a form of Wilcoxon test? If so, which one exactly?
10. Line 236: The authors should inform how many synthetic SMOTE samples were generated.
11. The Kaplan-Meier plots need some pimping: thinner lines, annotation of censored cases, number of remaining patients at given time intervals, ...
12. Line 343: Please refrain from statements like 'trend to significance'. The bar of $p=0.05$ is already low (i.e. accepting a 5% false positive result), and 0.249 is far from that bar.

13. Line 484: Please define 'variance'.
14. Why are the 2 cohorts combined for the tissue enrichment analysis? All other analyses are done on the separate cohorts.
15. Line 536: Why are 3 different experiment settings evaluated, and not just one (i.e. PDAC cancer vs. rest). Same question on line 85 (supplement).
16. Lab GitHub URL should be mentioned`
17. I'm surprised no patent filing is mentioned in the competing interests section. Why did the authors not consider to protect their findings?
18. It would be helpful if the authors could provide a scatterplot of fold changes of 39 DE genes in cohort 1 vs. 2 (PDAC vs. rest).

(Remarks on code availability)

I only scanned the GitHub page and based on the readme file, it seemed possible to redo the analyses and remake the figures.

Reviewer #3

(Remarks to the Author)

In the manuscript titled "Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-diagnostic High-risk and De-novo Symptomatic Patients" the authors describe a plasma cell-free RNA (cfRNA) evaluation of PDAC in pre-diagnostic high-risk and de novo symptomatic patients presenting for EUS-FNA. They use 153 blood samples for developing a cfRNA classifier and 95 samples for validation and report AUC of 0.966 in the discovery cohort and 0.889 in the independent validation cohort. The study addresses the need for developing accurate blood-based biomarkers for diagnosis of PDAC. The manuscript is well written and organized. The inclusion of cancer-free controls with pancreatic disease and the use of samples from two different centers are strengths of the study. The authors also describe an interesting approach for normalizing the molecular data. There are several limitations that warrant attention. These are summarized in comments below:

1. It is unclear if the primary aim of the study was to develop a blood-based biomarker for differentiating PDAC from benign pancreatic conditions versus a blood-based biomarker that determines individualized patient-level need for EUS. If the latter, the absence of blood samples from patients who did not undergo EUS as part of their diagnostic workup for similar indications is a limitation and should be discussed. In clinical practice there is a broad range of indications for EUS in pancreatic disease and it is unclear if all the EUS procedures performed for this study were to rule out cancer or a combination of different indications. Including the indication of EUS across the study population in both discovery and validation cohorts should be considered for interpretation of the results and to establish that the samples are representative of the intended use population of the test under development. The current study design, while demonstrating diagnostic performance of the cfRNA classifier in differentiating PDAC from a miscellaneous group of controls, does not support using it as a mechanism to determine the need for either EUS or EUS-FNA.
2. Please expand on the definition of the term 'benign pancreas' controls. Also, explain how this group is different from other cancer-free controls such as pancreatitis and IPMNs without advanced neoplasia in this cohort. Provide justification for not including a control group of patients without any pancreatic disease who are undergoing EUS for other non pancreatic indications.
3. The authors state "Additionally, EUS can identify small tumors that are not evident by other imaging modalities." Please add appropriate citations to support this statement, especially if there is data to support superiority in head-to-head comparisons with pancreas protocol CT and contrast-enhanced MRI. Although traditionally considered to be superior for PDAC detection, with recent advances in cross-sectional imaging, EUS for 'detection' appears to be comparable to other imaging modalities (PMID: 28624015).
4. The cfRNA biomarker candidates were highly expressed at potential metastatic sites of PDAC, especially liver. The authors state that this cfRNA analysis may detect micro-metastasis in the liver, which was not shown clinically by imaging. This statement can benefit from being supported with additional data. Consider including the distribution of metastatic sites for the PDAC cases in both discovery and validation cohorts and also clinical follow-up data on the PDAC cases without overt metastasis specifically describing the development of future metastatic disease. Also add group sizes (n=xx) to the x-axes of Figure 4a and 5a
5. In Figure 6 the authors indicate that the cfRNA classifier outperformed CA19-9 with both cross-validation in the CEDAR sample cohort and independent BCC validation cohort. Please explain how these results were compared, was CA 19-9 used as a continuous variable or at a particular threshold and also provide the confidence intervals around these AUC estimates or results of other tests of significance for this comparison either in the figure or text. Similarly describe how the combination of CA19-9 with cfRNA was compared with the classification accuracy of cfRNA and add insights for why the combination did not further enhance diagnostic performance. Moreover 10-20% of patients had no CA 19-9 value reported. Please describe why these were missing despite research protocol pre-procedure blood collection in all patients in the study and how the missing value was accounted for in the comparative analysis.
6. The authors highlight the pre-diagnostic collection of blood sample as a key strength of the study design. However, it is worthwhile noting that based on a paper by Pepe et al the PROBE approach has been considered a biomarker study design standard for over a decade. The PROBE approach clearly emphasizes prospectively collecting specimens and clinical data without need for diagnostic confirmation. This however does not preclude case-control comparisons which can be performed using a retrospective blinded nested case-control approach within prospectively collected biorepositories. The authors state that "In this study, blood was drawn before the EUS screening in high-risk and clinically indicated patients whose diagnoses were not confirmed". Does this mean that patients with an established diagnosis were excluded from the analysis in both case and control groups? A study flow diagram and description of how this study followed the PROBE approach should be included.
7. The PDAC score was correlated with stage which explains the outcomes in the survival analysis. The limited sample size

and stage dependence of the PDAC score make the survival analysis challenging to interpret without adjusting for stage and treatment. The figure summarizing this analysis may be moved to the supplemental section if needed.

8. Training a ML discrimination classifier on a small number of cases (n=20) with heterogeneous controls has inherent limitations and combined with the small sample size of cases (n=21) in the validation set remains a significant limitation. Also, limiting the study population only to patients referred for EUS potentially introduces bias. Many patients with pancreatitis and IPMNs will not have an indication for EUS in real world clinical settings. These limitations and how they impact the interpretation and future application of the study results should be summarized in the discussion of the manuscript.

9. Please review reference list and remove duplicates.

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed all of my concerns except for the following one:

When normalizing gene expression across all samples, including pancreatic and other cancer samples, will the genes specifically expressed in the pancreas (e.g., CTRB1/2) be removed or normalized along with the genes?

Response:

All genes in all samples are normalized together, we only remove those that have very low variance in the training cohort.

It needs to be very clear on the definition of "very low variance."

(Remarks on code availability)

The codes have improved and look good to me.

Reviewer #2

(Remarks to the Author)

The manuscript by Moore, Spiliotopoulos, Callahan et al. has substantially been revised and many comments have been satisfactorily addressed. However, a few remain and a few new have arisen.

1. The authors now mention the use of Streck tubes in the Supplemental material. However, there are several types of blood collection tubes offered by the Streck company. The authors should be specific (and ideally mention catalog number).
2. The authors use blood collection tubes that have an interval of 3 days between blood draw and plasma preparation. As others have shown, EDTA tubes and all Streck tubes are not suitable for RNA analyses for such long time intervals (e.g. <https://doi.org/10.1101/2021.05.11.442610> and others). The authors should comment on this (and recommend readers NOT to use plasma with such a long delay), and do primary QC of the plasma (e.g. quantify levels of hemolysis in the plasma). While the authors' method may be able to filter out genes that are variable because of processing, I want to avoid that the reader gets a wrong impression that it is okay to use plasma prepared with such a delay!
3. Figure legend S1 should explain what the centrifugation conditions SS1/2 and DS11/12 mean.
4. Reply to my major comment 3: It is still unclear from the text if cf-normalization includes TMM or not or if TMM is done after cf-normalization. It is also not clear if the novel cf-normalization is better than only classic normalization (using e.g. TMM).
5. Reply to my major comment 4: Supplementary Table 1 only contains details on the reference cohort. Where is the full clinical annotation of the patients? I see a Supplementary Data Table 1, but this one only contains the non-tumor cases and has no survival information?
6. Reply to my major comment 8: Why are CA19-9 and other known prognostic/predictive features not added to the Cox model? Why are two cohorts not combined to increase power? The authors do it for the tissue deconvolution.
7. The exonic read fraction is low for a set of samples (e.g. 13-60% for ~10% of samples), indicative of problematic DNA contamination. Did the authors exclude such samples?

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

I appreciate the authors detailed rebuttal to previous comments. While they have appropriately responded to most of the comments, listed below are a few additional queries and suggestions for modifications.

1. The authors acknowledge the limitation that the result from this study is not sufficient to determine the need for either EUS

or EUS-FNA. Hence, the last sentence in the abstract “Our findings could help to identify high-risk individuals for direction to an EUS-FNA procedure and improve its diagnostic utility” should be deleted as it is not supported by the data presented.

2. The authors indicate ‘benign pancreas’ group included pancreatic cysts. However, there was also a separate category of IPMNs which is a type of pancreatic cyst. It appears that there may be some overlap of disease states within the different subset of controls. If this is correct it should be addressed in the methods and discussion.

3. Speculative statement and discussion about detecting micro-metastasis in the liver should either be removed in the absence of data to support it or restated as a hypothesis that needs to be tested in future studies.

4. The authors state that 126 samples in the CEDAR cohort and 86 samples in the BCC cohort had a CA 19-9 value. Were the AUC comparisons between CA 19-9 and DE genes presented in Figure 6 limited to this subset of samples only.

Inclusion of sample size under disease categories on the axis in Figure 6a and 6b is recommended to improve clarity of results.

5. Unfortunately, I am unable to locate and review ‘Supplementary Data table 1’ in the review folder. Table 1 in the supplemental material describes sample handling conditions.

(Remarks on code availability)

NA

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

See attached document.

(Remarks on code availability)

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Point-by-point Response to Reviewers for

Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-diagnostic High-risk and *De-novo* Symptomatic Patients

Travis W. Moore^{*,1}, Elias Spiliotopoulos^{*,1}, Rowan Callahan^{*,1,2}, C. Ward Kirschbaum¹, Conner Bailey¹, Hyun Ji Kim^{1,2}, Breeshey Roskams-Hieter¹, Florian Goncalves^{1,2}, Dove Keith³, Aaron Grossberg^{1,3,4}, Paul T. Spellman¹, Gordon B. Mills⁵, Rosalie Sears^{3,6}, Terry K. Morgan^{1,2,7}, Thuy T. M. Ngo^{1,2,5,6,✉}

We would like to thank the reviewers for recognizing the significance and the novelty of our manuscript. The reviewers raised important concerns to be considered. We revised our manuscript to address all the concerns. The major changes are made as follows:

- We performed additional experiments and generated sequencing data of **additional 70 control samples** to identify the intrinsic and extrinsic factors of cell-free RNA (cfRNA). These 70 plasma samples were collected from 10 healthy individuals (5 male and 5 female donors). Each blood draw was processed with 7 different processing conditions with varying tube type (EDTA and Streck tubes), time from blood draw to processing (from 4 hours to 3 days), and centrifuge settings (single or double spins at various speeds). Results are added to the **NEW Supplementary Tables S1-2 and NEW Supplementary Figure S1**. The new sequencing data were uploaded to NCBI Sequencing Read Archive, along with sequencing data from patients.
- Based on the new sequencing data of 70 control samples, we improved the cell-free RNA normalization method to balance the expected intrinsic and extrinsic gene contributions in each sample. We updated the details of our **new cf-normalization algorithm in the Methods section**. The improved version of our cf-normalization method resulted in an increase in classification performance while maintaining the major findings of this study similar to the initial submission.
- We added **NEW Supplementary Data Table 1** to include metadata details of every patient including clinical EUS indications, clinical diagnosis of patients in the heterogeneous benign pancreas group, and metastasis sites. We added **NEW Supplementary Data Table 2** to include sequencing statistics of every patient.
- We **revised all the Results and Figures** using the improved version of our cfRNA normalization method.
- We addressed concerns related to (1) separating the discovery (CEDAR) and validation (BCC) cohorts in performing deconvolution (**NEW Supplementary Figure S7**); (2) the use of synthetic SMOTE samples (**NEW Supplementary Figure S10 and S11**); (3) comparing positive predictive value of our cfRNA PDAC score with CA19-9 (**NEW Supplementary Figure S17**), and (4) survival analysis with Cox proportional hazards model (**NEW Supplementary Table S5**).
- We revised our main text, method section, supplementary document, github code and instructions, to improve clarity, describe the new results, and discuss the limitations.

Our new data and analyses were consistent with our initial findings and further confirmed the main conclusion of the manuscript. We believe that our revised manuscript satisfactorily addressed all the questions raised by the reviewers.

Please see our point-by-point response to reviewers' concerns below (in blue):

Reviewer #1 (Remarks to the Author):

The authors identified 39 potential cfRNA biomarkers for the diagnosis of PDAC. They combined these biomarkers with a novel normalization method yielding discrimination accuracy for the separation of PDAC from all other intermediary pathologic conditions with an AUC of 0.966 by cross-validation in a discovery cohort and 0.889 in the independent validation cohort. The findings could help to identify high-risk individuals for direction to an EUS-FNA procedure thereby enhancing its diagnostic utility. The following are my concerns:

Line 128, "There were no statistically significant differences between the compositions of the sample sets (Table 1). In the PDAC groups, there were more males than females and the age range was higher than patients with other diagnoses ($P < 0.001$). ". It would be beneficial if the authors could list all statistical p values for compositions of the sample sets in Table 1. I wonder will the significant gender and age range imbalance impact all the follow up analyses in this study? Have your models being adjusted for gender and age range?

Response:

We added all statistical value in Revised Table 1.

Since the sample sets have sex imbalance, we tested the classification models for male and female separately (Revised Figure 4 & 5). The results are comparable with combining all samples of both sexes. To examine the impact of age range imbalance, we evaluated a binary split of patients younger or older than 68 (Revised Figure 4 & 5). We did not observe notable bias towards age.

L147, "To create this set of cfRNA intrinsic genes, we generated cfRNA sequencing data for plasma samples from three healthy individuals.". I wonder how did the author select the 3 healthy individuals? What the races are these 3 individuals? Moreover, the samples size here is very small and may not represent the broad healthy individuals to select the reference cfRNA genes for robust normalization.

Response:

We apologize for the limitation in sample set of healthy individuals we used to generate data for identifying cfRNA intrinsic genes in the initial submission. In this submission, we generated additional cfRNA sequencing data for new 70 plasma samples from 10 healthy individuals (5 male and 5 female donors). Each blood draw was processed with 7 different processing conditions with varying tube type (EDTA and Streck tubes), time from blood draw to processing (from 4 hours to 3 days), and centrifuge settings (single or double spins with various speed). Details of the 7 processing conditions are provided in new supplementary table S1. New data are presented in new Supplementary Figure S1, and supplementary table S1-2.

L235, "To address the large imbalance between these two classes in the data, we used the SMOTE interpolation method [47] to create synthetic PDAC samples for training. " I am

concerned about the synthetic PDAC samples for training, how many synthetic PDAC samples were generated for training? Will the synthetic PDAC samples being too ideal and significantly different with the real PDAC samples? Moreover, it would be helpful to perform the training without synthetic PDAC samples as a comparison.

[Response:](#)

100 synthetic PDAC samples were generated for training by SMOTE interpolation method. We compared SMOTE PDAC samples with real PDAC using PCA and UMAP and observed no difference (New supplementary Figure S10). Following the reviewer's suggestion, we performed the training without synthetic PDAC samples (New supplementary Figure S11).

When normalizing gene expression across all samples, including pancreatic and other cancer samples, will the genes specifically expressed in the pancreas (e.g., CTRB1/2) be removed or normalized along with the genes?

[Response:](#)

All genes in all samples are normalized together, we only remove those that have very low variance in the training cohort.

Reviewer #1 (Remarks on code availability):

To make sure the paper are reproducible and the code is a reusable resource for the community, the authors need to add the tutorial/manual on how to use the codes, for instance, 2024_ngolab_cfRNA/Analysis_Scripts/DE_wilcox.R, what is the format of input file, the commands on how to use the codes, what is the output and how to interpret them?

[Response:](#)

We revised the detailed instruction and input files on github for re-using the code.

Reviewer #2 (Remarks to the Author):

The manuscript entitled "Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-diagnostic High-risk and De-novo Symptomatic Patients" by Moore, Spiliotopoulos, Callahan et al. reports on a method and a 39 cell-free RNA gene classifier to diagnose pancreatic ductal adenocarcinoma in a pre-diagnostic high-risk de novo symptomatic patient populations. The manuscript is well written, the figures are illustrative, the results support the conclusions, and the study has a promising impact for better future clinical management of patients. Nevertheless, I have a series of technical questions and comments to ascertain the validity of the results.

major comments:

1. The blood collection tube type and time between blood draw and plasma preparation are two important variables that impact the extracellular RNA repertoire. These variables, as well as others (e.g., fasting status at the time of blood draw, etc.), are not mentioned in the study but

are important for replicating the study and evaluating potential confounding variables. According to recent guidelines, the authors should report pre-analytical variables, e.g., CEN/TS 17742:2022 ("Molecular in vitro diagnostic examinations – specifications for pre-examination processes for venous whole blood – isolated circulating cell free RNA from plasma").

Response:

We fully agree with the reviewer that tube type, time between blood draw and plasma preparation are important parameters to control. Therefore, we generated new data from additional 70 plasma samples from 10 healthy individuals (5 male and 5 female donors). Each blood draw was processed with 7 different processing conditions with varying tube type (EDTA and Streck tubes), time from blood draw to processing (from 4 hours to 3 days), and centrifuge settings (single or double spins with various speed). We highlighted the difference in cfRNA sequencing data of these samples (New Supplementary Figure S1). We used this data set to perform the analysis to determine a new normalization factor for each gene. We reanalyzed all models using this updated cfRNA normalization method. The results are comparable to the report in the initial submission.

We revised the method section to highlight the details of the blood collection method. Blood specimens were collected in EDTA tubes before the clinical procedure. We do not control for time of blood draw and fasting status other than what were required in the standard clinical visit. We also noted plasma process condition in the method section. Patient plasma was processed from whole blood by centrifugation twice; first at 1,000g for 10 minutes, then the supernatant plasma was removed and spun at 2,500g for 10 minutes at room temperature. Plasma aliquots were frozen and stored at -80°C until we began RNA processing

2. While the authors make a laudable effort to remove variable genes that result from variable platelet content (because of different centrifugation protocols), I'm surprised they only looked at centrifugation to determine so-called batch effects and improve the normalization. It is well-known that the time between blood draw and plasma preparation is likely an even more important factor, which the authors did not evaluate here. This may be acceptable if this pre-analytical variable was controlled well in this study, but that information is not provided (see previous point). Anyhow, this variable's impact should at least be discussed and serve as a warning for the reader. The same is true for the level of hemolysis, the temperature during blood storage/transport before plasma preparation, etc.

Response:

As suggested by the reviewers, in this revision we generated new data from additional 70 plasma samples from 10 healthy individuals (5 male and 5 female donors). Each blood draw was processed with 7 different processing conditions with varying tube type (EDTA and Streck tubes), time from blood draw to processing (from 4 hours to 3 days), and centrifuge settings (single or double spins with various speed). We used this data set to perform the analysis to determine a new normalization method. (See previous response point).

We added the intrinsic and extrinsic factor for a list of selected platelet genes in the Supplementary document.

3. This reviewer thought that you could only do one normalization procedure (defined as applying a sample-specific factor to remove technical variability based on sample-specific gene expression values). Therefore, I'm confused by the authors' application of their intrinsic normalization procedure followed by TMM normalization. Can the authors confirm that intrinsic followed by TMM differs from doing TMM only (when using the same set of genes)?

Response:

The reviewer is correct that normalization involves applying sample-specific correction factors that affect all genes in that sample. The factors computed by TMM depend on the initial variance of the data, as it trims extreme gene values in samples when calculating means to reduce the influence of outliers. Applying a different normalization first can change the variance of the data, causing TMM to trim different genes and produce a different correction then if run on the original data. That said, our new normalization method adjusts individual gene levels differently in each sample, rather than adjusting all genes by a sample specific factor.

4. The raw sequencing data and data-processing and analysis scripts are publicly available. But is the patient annotation file also available (with matched identifiers between the raw data and the annotation)? Also, key intermediate result matrices should also be made available on the GitHub, like raw and normalized gene counts.

Response:

Raw sequencing data are deposited on Sequencing Read Archive. Data processing and analysis scripts are available on Github. We included annotation of all samples in the Supplementary Table 1. Raw and normalized gene counts are available on GitHub.

5. ROC AUC values indicate that the measurements have classification potential but are not very informative for the individual patient. The authors are invited to mention positive and negative predictive values (like they do for the CA19-9 biomarker).

Response:

As suggested, we added positive and predictive values (New Supplementary Figure 17).

6. I'm well aware of (often ignored or underappreciated) batch effects in RNA seq studies, resulting from RNA extraction and/or library preparation and/or sequencing lane/flow cell. It would be good if the authors could annotate their samples in terms of at least these potential batch effects. For RNA extraction, the authors mention randomization (to avoid batch effects), but what about library prep and sequencing? As an example, did all samples receive a unique index so all could be pooled and sequenced on the same set of sequencing lanes?

Response:

We randomized all samples for RNA extraction, library preparation and sequencing. All samples have unique index and were pooled and randomly assigned to the same sequencing lanes. We added these details in the revised method section.

7. Given the pervasive nature of batch effects, I'm surprised that a sequencing-based classifier works so well when trained on one cohort/batch and tested on another one. Could the authors clarify how they ensured that the normalized results have the same scale and variance, assuming a RF classifier is sensitive to scale and variance? Simply following the same data processing pipeline is likely not going to result in comparable datasets. Did the authors ensure that the sequencing depth was the same between the 2 cohorts, was some additional data rescaling done after normalization, ...?

Response:

Our normalization procedure ensures that all samples have comparable overall sequencing depth. Classifiers are passed the log transformed normalized gene count matrices to further reduce outlier influence. We have added this detail to the methodology.

8. The use of a Kaplan-Meier plot forces the authors to define a cut-off that may not be relevant or correct. I suggest they do Cox proportional hazard modeling of the score instead, either with or without the inclusion of additional variables (like stage 1-2 vs 3-4, CA19-9, ...).

Response:

We have added the output of a Cox proportional hazard model to the supplemental material. We have also added error bars to the Kaplan-Meier plots.

9. The methods section should provide more details: how much plasma was used as input for RNA extraction, the RNA elution volume, how much RNA was DNase treated, the RNA eluate volume after the second purification, library QC, library concentration measurement, library loading concentration on the sequencer, read length, how many genes were removed in the variance filter, how many remained after that filter, 'divided by the total expression of the top 100 intrinsic genes' (how is this done, what is 'total expression'), 'normalized across tissues using z-scores' (how is this done, is this really normalization?), software version are lacking, etc.

Response:

We revised the method section to include all the details: plasma volume, RNA elution volume in each step, DNA digestion, libraries preparation input, library QC and concentration, read length and sequencing depth. After sequencing processing, the library sizes and RNA distribution are included in the New Supplementary Data Table 2.

Z-scores of atlas tissue enrichments for DE genes are computed on a per-gene level across tissues, by subtracting the mean and dividing by the standard deviation. This gives a consistent scale to compare relative enrichments across genes. These details have been added to the methodology.

minor comments:

1. Figure 1: What is benign pancreas? Is this a normal pancreas, or benign pancreatic cysts, or something else?

Response:

We added New Supplementary Data Table 1 to include details of each benign pancreas patients. There is a wide range of different disease conditions we include in this groups, such as no mass lesion detected, bile duct stones, pancreatic cysts, benign reactive changes, etc.

2. Line 140: Is 'batch' factor a good term? Plasma preparation differences mainly impact an individual sample, while 'batch' is a term that is typically used for systematic technical differences introduced for a set of samples that are processed together.

Response:

We have re-defined this as 'extrinsic' factor in the text and figures, to emphasize that it is a sample specific effect that occurs after blood draw.

3. The batch factor is likely driven entirely by platelet RNA content. Can the authors demonstrate enrichment of platelet signal in the 'batch' genes?

Response:

Our data supports reviewer's speculation. We added a list of selected platelet genes with their corresponding intrinsic and extrinsic factor to demonstrate enrichment of platelet signal in the extrinsic factor (supplemental table S3). We will additionally include the intrinsic and extrinsic factor values in our github.

4. Please replace all instances of housekeeping gene by reference gene.

Response:

We have replaced all instances of housekeeping gene by reference gene in this revision.

5. Line 157: What is 'the total gene expression'?

Response:

We have replaced 'the total gene expression' by the library size.

6. As the authors measure extracellular RNA in blood plasma, the term 'expression' is not very well suited. Plasma does not express any transcript; instead, transcript are more or less (differentially) abundant, or show differences in concentration (but not in expression levels). Expression is a term better reserved for cellular RNA, as part of an active process.

Response:

We agree with the review and thanks for catching this terminology error. We replace the term “expression” by count or level for cell-free RNA.

7. The list of 100 intrinsic genes along with their TPM values should be given in a supplementary file.

Response:

Our updated normalization uses learned factor values for each gene. We have included these factor values in our github repository.

8. Line 181: It is unclear if the average depth of ~1 million reads per sample is after deduplication.

Response:

Yes, average 1 million reads per sample is after deduplication.

9. Please be more precise about the statistical test used to identify the differentially expressed genes. A Wilcox test is not a test. Do the authors mean a form of Wilcoxon test? If so, which one exactly?

Response:

We thank the reviewer for catching the typo. We corrected it to Wilcoxon rank sum.

10. Line 236: The authors should inform how many synthetic SMOTE samples were generated.

Response:

100 SMOTE samples were added to training data, which we have made explicit in the main text and supplemental.

11. The Kaplan-Meier plots need some pimping: thinner lines, annotation of censored cases, number of remaining patients at given time intervals, ...

Response:

We have revised Supplementary Figure S18 following reviewer's suggestion.

12. Line 343: Please refrain from statements like ‘trend to significance’. The bar of $p=0.05$ is already low (i.e. accepting a 5% false positive result), and 0.249 is far from that bar.

Response:

We have removed this language, and indicate throughout the paper when p-values are significant or not.

13. Line 484: Please define 'variance'.

[Response:](#)

Estimated sample variance is the variance calculated in this step, which we have corrected in the text.

14. Why are the 2 cohorts combined for the tissue enrichment analysis? All other analyses are done on the separate cohorts.

[Response:](#)

Tissue deconvolution results are performed on the combined cohorts to increase the number of PDAC samples in the analysis. We have additionally included deconvolutions of the cohorts separately in Supplementary Figure S7.

15. Line 536: Why are 3 different experiment settings evaluated, and not just one (i.e. PDAC cancer vs. rest). Same question on line 85 (supplement).

[Response:](#)

We perform three different experiments to show the results of our classification of PDAC under increasingly more heterogeneous populations. Of note, the inclusion of other cancer types causes a decrease in classifier performance while the inclusion of intermediate pathologies does not.

16. Lab GitHub URL should be mentioned`

[Response:](#)

GitHub URL was included.

17. I'm surprised no patent filing is mentioned in the competing interests section. Why did the authors not consider to protect their findings?

[Response:](#)

We did not pursue IP protection due to limited funding for IP filing at our institution.

18. It would be helpful if the authors could provide a scatterplot of fold changes of 39 DE genes in cohort 1 vs. 2 (PDAC vs. rest).

[Response:](#)

We have included a volcano plot of DE gene discovery for our training cohort in supplemental figure 4. Boxplots of DE gene values in both cohorts are in supplementary figures S5 and S6.

Reviewer #2 (Remarks on code availability):

I only scanned the GitHub page and based on the readme file, it seemed possible to redo the analyses and remake the figures.

[Response:](#)

[We included all the analysis code and the instruction on GitHub.](#)

Reviewer #3 (Remarks to the Author):

In the manuscript titled “Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-diagnostic High-risk and De-novo Symptomatic Patients” the authors describe a plasma cell-free RNA (cfRNA) evaluation of PDAC in pre-diagnostic high-risk and de novo symptomatic patients presenting for EUS-FNA. They use 153 blood samples for developing a cfRNA classifier and 95 samples for validation and report AUC of 0.966 in the discovery cohort and 0.889 in the independent validation cohort. The study addresses the need for developing accurate blood-based biomarkers for diagnosis of PDAC. The manuscript is well written and organized. The inclusion of cancer-free controls with pancreatic disease and the use of samples from two different centers are strengths of the study. The authors also describe an interesting approach for normalizing the molecular data. There are several limitations that warrant attention. These are summarized in comments below:

1. It is unclear if the primary aim of the study was to develop a blood-based biomarker for differentiating PDAC from benign pancreatic conditions versus a blood-based biomarker that determines individualized patient-level need for EUS. If the latter, the absence of blood samples from patients who did not undergo EUS as part of their diagnostic workup for similar indications is a limitation and should be discussed. In clinical practice there is a broad range of indications for EUS in pancreatic disease and it is unclear if all the EUS procedures performed for this study were to rule out cancer or a combination of different indications. Including the indication of EUS across the study population in both discovery and validation cohorts should be considered for interpretation of the results and to establish that the samples are representative of the intended use population of the test under development. The current study design, while demonstrating diagnostic performance of the cfRNA classifier in differentiating PDAC from a miscellaneous group of controls, does not support using it as a mechanism to determine the need for either EUS or EUS-FNA.

[Response:](#)

[The primary aim of this study was to develop blood-based biomarkers using cell-free RNA for differentiating PDAC from various pancreas related diseases with intermediary pathologic conditions that complicate the diagnosis of PDAC in a clinical setting. We chose to recruit patients who were referred to the EUS procedures because: \(1\) This group of patients include individuals who have high-risk for PDAC such as family history and genetic predisposition and patients with symptoms, medical history and imaging indicated PDAC diseases and cancer; \(2\) This clinical setting allows us to effectively and practically \(with limited academic research](#)

resource) collect samples before definitive diagnosis while enriching for the proportion of PDAC outcome.

There is a broad range of clinical indications for EUS to be performed for patients included on our study. We included details of EUS indication for every patient included in this study in the Supplementary Data Table 1.

We acknowledge the limitation that the results from this study is not sufficient to determine the need for either EUS or EUS-FNA. We revised the Discussion Section to include this limitation.

2. Please expand on the definition of the term 'benign pancreas' controls. Also, explain how this group is different from other cancer-free controls such as pancreatitis and IPMNs without advanced neoplasia in this cohort. Provide justification for not including a control group of patients without any pancreatic disease who are undergoing EUS for other non pancreatic indications.

Response:

The "benign pancreas" group include patients with various diseases and benign conditions such as no mass lesion detected, bile duct stones, pancreatic cysts, benign reactive changes, etc. We revised the text and added New Supplementary Data Table 1 to note the details of each benign pancreas patients. Patients without any pancreatic disease who are undergoing EUS for other non-pancreatic indications were grouped together within the "benign pancreas" group.

3. The authors state "Additionally, EUS can identify small tumors that are not evident by other imaging modalities." Please add appropriate citations to support this statement, especially if there is data to support superiority in head-to-head comparisons with pancreas protocol CT and contrast-enhanced MRI. Although traditionally considered to be superior for PDAC detection, with recent advances in cross-sectional imaging, EUS for 'detection' appears to be comparable to other imaging modalities (PMID: 28624015).

Response:

To support the sentence "Additionally, EUS can identify small tumors that are not evident by other imaging modalities", we added the reference PMID 30406288: "M. Kitano, T. Yoshida, M. Itonaga, T. Tamura, K. Hatamaru and Y. Yamashita, "Impact of endoscopic ultrasonography on diagnosis of pancreatic cancer," vol. 54, pp. 19-32, 2019".

4. The cfRNA biomarker candidates were highly expressed at potential metastatic sites of PDAC, especially liver. The authors state that this cfRNA analysis may detect micro-metastasis in the liver, which was not shown clinically by imaging. This statement can benefit from being supported with additional data. Consider including the distribution of metastatic sites for the PDAC cases in both discovery and validation cohorts and also clinical follow-up data on the PDAC cases without overt metastasis specifically describing the development of future metastatic disease. Also add group sizes (n=xx) to the x-axes of Figure 4a and 5a

Response:

We added New Supplementary Data Table 1 with a column to indicate metastasis sites of every PDAC patients in both discovery and validation cohorts. Unfortunately, we do not have clinical follow-up data on all PDAC cases because some patients were referral patients and did not come back to our hospital for follow-up. Because we do not have sufficient data to support the statement, we speculated that “This observation indicates that our cfRNA analysis may detect micro-metastasis in the liver, which was not shown clinically by imaging, or that the liver may be involved and remodeled early during PDAC carcinogenesis and progression potentially by circulating factors from the PDAC”.

5. In Figure 6 the authors indicate that the cfRNA classifier outperformed CA19-9 with both cross-validation in the CEDAR sample cohort and independent BCC validation cohort. Please explain how these results were compared, was CA 19-9 used as a continuous variable or at a particular threshold and also provide the confidence intervals around these AUC estimates or results of other tests of significance for this comparison either in the figure or text. Similarly describe how the combination of CA19-9 with cfRNA was compared with the classification accuracy of cfRNA and add insights for why the combination did not further enhance diagnostic performance. Moreover 10-20% of patients had no CA 19-9 value reported. Please describe why these were missing despite research protocol pre-procedure blood collection in all patients in the study and how the missing value was accounted for in the comparative analysis.

Response:

Classifier performance of CA 19-9 was evaluated using CA 19-9 levels as a continuous variable, either by itself or with our detected DE genes. We have included significance test outcomes between CA 19-9 AUC values and our DE gene classifier, and indicated the correlation of CA 19-9 and several DE genes as a potential explanation why CA 19-9 did not improve classification when added to DE gene features.

Not all of the patients in this study had CA 19-9 value near the time of blood draw in our electronic medical record database. Partly, it is because many patients are referral patients from other healthcare centers. In addition, we do not have sufficient serum volume left from the same blood draw to retrospectively perform CA 19-9 measurement for a fraction of patients. We did not specifically request and review manual medical record transfer for this study.

6. The authors highlight the pre-diagnostic collection of blood sample as a key strength of the study design. However, it is worthwhile noting that based on a paper by Pepe et al the PRoBE approach has been considered a biomarker study design standard for over a decade. The PRoBE approach clearly emphasizes prospectively collecting specimens and clinical data without need for diagnostic confirmation. This however does not preclude case-control comparisons which can be performed using a retrospective blinded nested case-control approach within prospectively collected biorepositories. The authors state that “In this study, blood was drawn before the EUS screening in high-risk and clinically indicated patients whose diagnoses were not confirmed”. Does this mean that patients with an established diagnosis were excluded from the analysis in both case and control groups? A study flow diagram and description of how this study followed the PRoBE approach should be included.

Response:

We thank the reviewer for noting the P_{Ro}BE approach by Pepe et al. We have added this reference in the introduction.

We included patients with blood draw performed before the EUS procedure. We did not include patients with established diagnosis. In the new Supplementary Data Table 1, we listed EUS indication for every patient.

We presented the study diagram in the Figure 1. We did not strictly design the P_{Ro}BE approach specifically for this study. All the samples were collected and banked by Brenden-Colson Center for Pancreatic Care (BCC) and Cancer Early Detection Advanced Research Center (CEDAR). Banked samples were released to multiple studies. Upon our request, samples with sufficient plasma volume were released to perform this study. We do not have details of the number of patients included in each step of patient recruitment and sample collection process (patient identification, selection, reach out, recruitment, specimen collection, and outcomes). We revised our method section to clarify the sample collection, banking, and releasing process.

7. The PDAC score was correlated with stage which explains the outcomes in the survival analysis. The limited sample size and stage dependence of the PDAC score make the survival analysis challenging to interpret without adjusting for stage and treatment. The figure summarizing this analysis may be moved to the supplemental section if needed.

Response:

We have moved survival result figures to the supplementary document.

8. Training a ML discrimination classifier on a small number of cases (n=20) with heterogeneous controls has inherent limitations and combined with the small sample size of cases (n=21) in the validation set remains a significant limitation. Also, limiting the study population only to patients referred for EUS potentially introduces bias. Many patients with pancreatitis and IPMNs will not have an indication for EUS in real world clinical settings. These limitations and how they impact the interpretation and future application of the study results should be summarized in the discussion of the manuscript.

Response:

We thank the reviewer for this suggestion. We added a paragraph in the discussion section to acknowledge the significant limitation of the small sample size the potential bias introduced toward the patients with EUS indication only. We noted that whether our cfRNA approach reported in the current study is applicable for patients with pancreatitis and IPMN without EUS indication remains to be explored in future studies.

9. Please review reference list and remove duplicates.

Response:

Thank you, we have removed the duplicate citation.

Point-by-point Response to Reviewers for the Revision of
Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-diagnostic High-risk and *De-novo* Symptomatic Patients

Travis W. Moore^{*,1}, Elias Spiliotopoulos^{*,1}, Rowan Callahan^{*,1,2}, C. Ward Kirschbaum¹, Conner Bailey¹, Hyun Ji Kim^{1,2}, Breeshey Roskams-Hieter¹, Florian Goncalves^{1,2}, Dove Keith³, Aaron Grossberg^{1,3,4}, Paul T. Spellman¹, Gordon B. Mills⁵, Rosalie Sears^{3,6}, Terry K. Morgan^{1,2,7}, Thuy T. M. Ngo^{1,2,5,6,✉}

We would like to thank the reviewers for their valuable feedback in this second round of revision. We have revised our manuscript to address all the concerns. The changes are made as followed:

- We added additional analysis with TMM normalization for comparison. TMM normalization showed a poorer performance compared to our normalization method.
- We also added additional classification analysis after filtering samples with exon fraction cut off at 0.6 for comparison. There is no significant change compared to our original analysis.
- We also revised our main text, method section, supplementary document, to improve clarity. We added details to ensure all clinical annotation information is clearly included in the Supplementary Data Table 1 as a separate excel file.
- We added additional survival analysis combining samples in both cohorts to increase the number of samples.

Please see our point-by-point response to reviewers' concerns below (in blue):

REVIEWER COMMENTS

Reviewer #1 (Remarks to the Author):

The authors have addressed all of my concerns except for the following one:

When normalizing gene expression across all samples, including pancreatic and other cancer samples, will the genes specifically expressed in the pancreas (e.g., CTRB1/2) be removed or normalized along with the genes?

Response:

All genes in all samples are normalized together, we only remove those that have very low variance in the training cohort.

It needs to be very clear on the definition of "very low variance.".

Response: We apologize for a brief response in the previous revision. Filtering is performed for genes that have a sample variance of less than 1, as computed on the normalized count matrices. This step was described in our methodology, under the section "DE Gene Discovery".

Reviewer #1 (Remarks on code availability):

The codes have improved and look good to me.

Reviewer #2 (Remarks to the Author):

The manuscript by Moore, Spiliotopoulos, Callahan et al. has substantially been revised and many comments have been satisfactorily addressed. However, a few remain and a few new have arisen.

1. The authors now mention the use of Streck tubes in the Supplemental material. However, there are several types of blood collection tubes offered by the Streck company. The authors should be specific (and ideally mention catalog number).

Response: We used Streck RNA complete BCT tubes. This information is added to the Supplementary Table S1.

2. The authors use blood collection tubes that have an interval of 3 days between blood draw and plasma preparation. As others have shown, EDTA tubes and all Streck tubes are not suitable for RNA analyses for such long time intervals (e.g. <https://doi.org/10.1101/2021.05.11.442610> and others). The authors should comment on this (and recommend readers NOT to use plasma with such a long delay), and do primary QC of the plasma (e.g. quantify levels of hemolysis in the plasma). While the authors' method may be able to filter out genes that are variable because of processing, I want to avoid that the reader gets a wrong impression that it is okay to use plasma prepared with such a delay!

Response: Following reviewer's suggestion, we added a note in the discussion section that the various processing conditions were only used to identify the intrinsic and extrinsic factor for developing the normalization method. The processing condition for actual patient blood collection were uniform without extended delay as described in the method section.

3. Figure legend S1 should explain what the centrifugation conditions SS1/2 and DS11/12 mean.

Response: We have updated the figure to specify the spin speeds for each condition.

4. Reply to my major comment 3: It is still unclear from the text if cf-normalization includes TMM or not or if TMM is done after cf-normalization. It is also not clear if the novel cf-normalization is better than only classic normalization (using e.g. TMM).

Response: To clarify, our cf-normalization as outlined in the paper is a separate procedure from TMM, and we use a normalization pipeline of cf-normalization followed by TMM for the samples in our analysis. We have added results using only TMM to the supplemental, as a baseline for comparison along with TPM. TMM was originally omitted as a baseline because it had inferior performance to TPM.

5. Reply to my major comment 4: Supplementary Table 1 only contains details on the reference cohort. Where is the full clinical annotation of the patients? I see a Supplementary Data Table 1, but this one only contains the non-tumor cases and has no survival information?

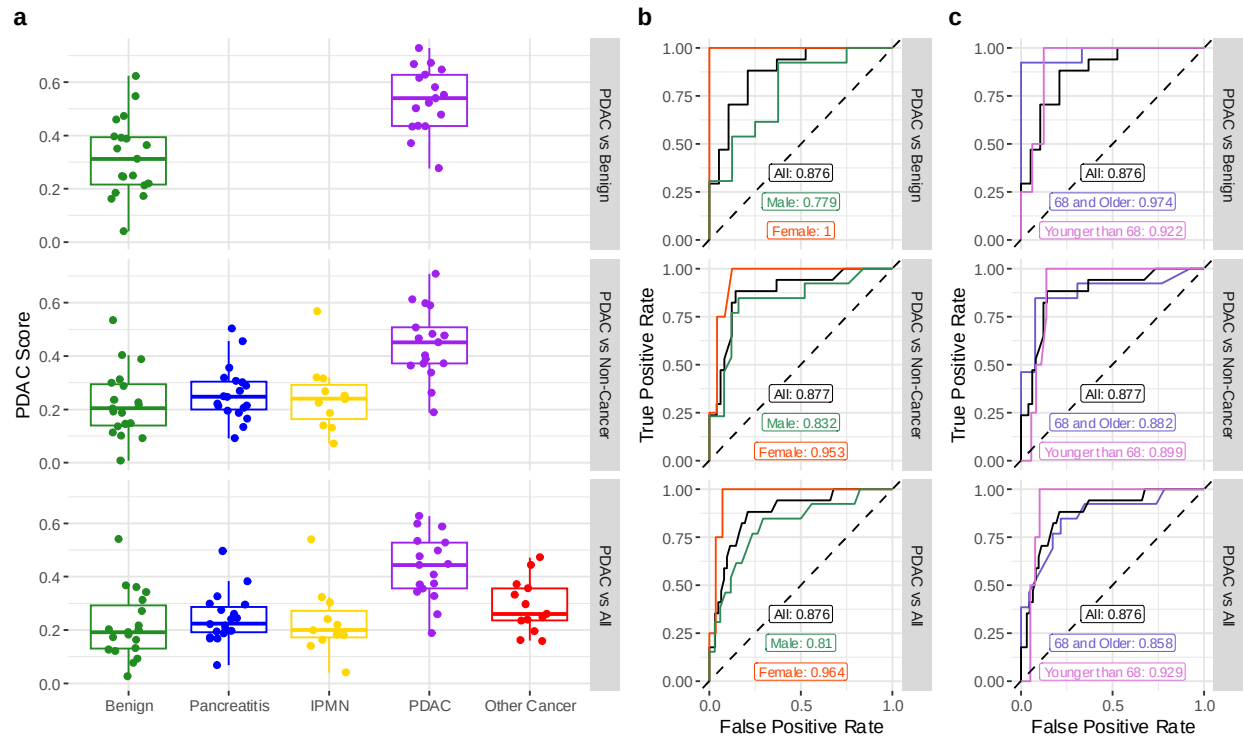
Response: Full clinical annotation of ALL the patients of the CEDAR and BCC cohorts is detailed in a separate excel file named "Supplementary Data Table 1". The Supplementary Table S1 within the document file named "Supplemental Material" is a different table and only contain information of the healthy control samples that we used for various blood processing conditions to develop the normalization method.

6. Reply to my major comment 8: Why are CA19-9 and other known prognostic/predictive features not added to the Cox mode? Why are two cohorts not combined to increase power? The authors do it for the tissue deconvolution.

Response: We have taken the reviewer's advice and combined the cohorts for the survival analysis, to increase power. We have added PDAC disease stage as another feature to the Cox model, which shows the highest significance. CA19-9 measurements were not available for 13 of the PDAC samples, so we have not included it to prevent from having to omit those samples from analysis.

7. The exonic read fraction is low for a set of samples (e.g. 13-60% for ~10% of samples), indicative of problematic DNA contamination. Did the authors exclude such samples?

Response: To retain the maximum number of samples, we did not exclude any samples based on exon read fraction. To address reviewer concern about samples with low exon fractions, we performed additional analysis for samples with exon fraction of more than 0.6 only. The training and validation AUC performance of this new analysis with exon filter cutoff of 0.6 is largely the same as the original analysis, though more DE genes are found when the filter is used. We have added a supplementary figure and table showing results when using an exon filter cutoff of 0.6.



Supplementary Figure S18. Validation results when removing samples with an exon fraction of less than 0.6.

Cutoff Value	Samples Removed: Training			Samples Removed: Validation			DE Genes		AUC PDAC vs All	
	<i>PDAC</i>	<i>Other</i>	<i>Total</i>	<i>PDAC</i>	<i>Other</i>	<i>Total</i>	<i>Total</i>	<i>Overlap</i>	<i>Training</i>	<i>Validation</i>
None	0	0	0	0	0	0	29	--	0.956	0.896
0.60	1	3	4	4	12	16	41	25	0.954	0.876

Supplementary Table S5. Exon filtering results. When a filter of 0.6 is applied, 41 DE genes are discovered from the Training data, though 25 of these are the same as the DE gene set found without any filtering. Training and validation performance is largely the same between the two.

Reviewer #3 (Remarks to the Author):

I appreciate the authors detailed rebuttal to previous comments. While they have appropriately responded to most of the comments, listed below are a few additional queries and suggestions for modifications.

1. The authors acknowledge the limitation that the result from this study is not sufficient to determine the need for either EUS or EUS-FNA. Hence, the last sentence in the abstract “Our findings could help to identify high-risk individuals for direction to an EUS-FNA procedure and improve its diagnostic utility” should be deleted as it is not supported by the data presented.

Response: As suggested, we have removed this sentence from the abstract.

2. The authors indicate ‘benign pancreas’ group included pancreatic cysts. However, there was

also a separate category of IPMNs which is a type of pancreatic cyst. It appears that there may be some overlap of disease states within the different subset of controls. If this is correct it should be addressed in the methods and discussion.

Response: We do not have overlap in cyst types. Diagnostic annotation was performed by a board-certified pathologist at Oregon Health and Science University based on their chart review and review of histologic sections. Our pathologist classified cystic lesions into either benign (serous cysts) or dysplastic intraductal papillary mucinous neoplasia (IPMN), which has malignant potential.

3. Speculative statement and discussion about detecting micro-metastasis in the liver should either be removed in the absence of data to support it or restated as a hypothesis that needs to be tested in future studies.

Response: We have removed this statement and discussion.

4. The authors state that 126 samples in the CEDAR cohort and 86 samples in the BCC cohort had a CA 19-9 value. Were the AUC comparisons between CA 19-9 and DE genes presented in Figure 6 limited to this subset of samples only. Inclusion of sample size under disease categories on the axis in Figure 6a and 6b is recommended to improve clarity of results.

Response: The reviewer is correct; Figure 6 only includes the subset of samples with CA 19-9 values. We have modified the figure to show the number of samples available in each disease category, as suggested.

5. Unfortunately, I am unable to locate and review 'Supplementary Data table 1' in the review folder. Table 1 in the supplemental material describes sample handling conditions.

Response: We make sure to include the separate excel file "Supplementary Data Table 1" which contain a full clinical annotation for all patients in both CEDAR and BCC cohorts.

Reviewer #3 (Remarks on code availability):

NA

Response to Reviewers for the Final Revision of
Enabling Non-invasive Detection of Pancreatic Ductal Adenocarcinoma in Pre-
diagnostic High-risk and *De-novo* Symptomatic Patients

Travis W. Moore^{*,1}, Elias Spiliotopoulos^{*,1}, Rowan Callahan^{*,1,2}, C. Ward Kirschbaum¹,
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Morgan^{1,2,7}, Thuy T. M. Ngo^{1,2,5,6},✉

We thank the reviewers for their additional review of our revised manuscript. We have corrected the error in the text and added clarifications to address all of the reviewers' concerns.

Reviewer Comments

Reviewer 2

1. CPM per factor

Based on the formula (CPM(intrinsic) + CPM(extrinsic)), I would expect a sum of around 2M instead of approximately 1M per sample in the middle plot below.

For normalization, according to the GitHub script, the factors are indeed divided by the sum but then multiplied by the median of the sum within a particular factor (instead of multiplying by 1M to obtain CPM). Thus, the multiplication factor could differ between intrinsic and extrinsic factors. Is this an older version of the script, or is the formula stated in the paper incorrect?

Response: Thank you for catching the outdated text in the method section. We have updated the text to reflect this change. After being split into the two factors, the counts matrices are normalized to the median column sum, giving a different normalization factor for intrinsic and extrinsic.

2. Estimation of factors based on reference:

The authors derive intrinsic and extrinsic factors per gene and perform refitting for genes of interest not present in the reference. I suspect the authors don't intend to present their derived values as complete and absolute. Additional genes will likely be missing for other

experiments, and perhaps another reference may be necessary. The practical implementation is not entirely clear. On GitHub, there is indeed a script for final normalization based on the derived factors, but regarding the factors themselves, I only found a table with values (no script explaining how these values were obtained).

Response: We have added a note to the methods to clarify that this approach of extending the learned factors to new genes is an estimate when reference values are not available. We have added the scripts for learning these factors from reference data, and extending them to new genes, to our GitHub repository.

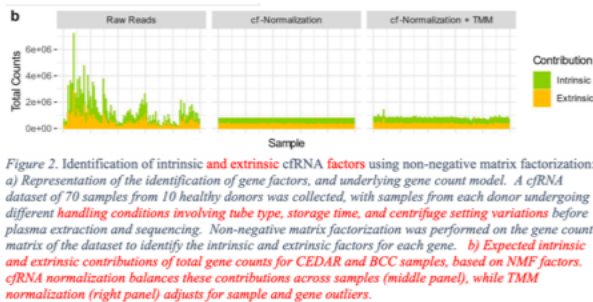
Reviewer 2 Review Attachment 1

The authors have addressed all my remaining concerns. However, when reviewing the code, I noticed the following. Please accept my apologies for not asking this before to the authors.

1. CPM per factor

sample j and gene i , where $I_i^n = I_i / (I_i + E_i)$ and $E_i^n = E_i / (I_i + E_i)$. Both $I_i^n g_{i,j}$ and $E_i^n g_{i,j}$ undergo CPM normalization before being combined back into the cf-normalized gene table $g_{i,j}^n = \text{cpm}(I_i^n g_{i,j}) + \text{cpm}(E_i^n g_{i,j})$. Performing CPM normalization on these contributions separately ensures that the total

Based on the formula (CPM(intrinsic) + CPM(extrinsic)), I would expect a sum of around 2M instead of approximately 1M per sample in the middle plot below.



For normalization, according to the GitHub script, the factors are indeed divided by the sum but then multiplied by the median of the sum within a particular factor (instead of multiplying by 1M to obtain CPM). Thus, the multiplication factor could differ between intrinsic and extrinsic factors. Is this an older version of the script, or is the formula stated in the paper incorrect?

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